

PC5202, Advanced Statistical Mechanics

Midterm Test, 9 March 2013

1. Consider thermodynamic problems.
 - a. Derive the Gibbs-Duhem relation for fluid system.
 - b. Use the result above to prove the Clapeyron equation, $dP/dT = \ell/(T \Delta v)$, for the slope of the phase-coexistence curve in a P - T diagram, where P is pressure, and T is absolute temperature. Interpret the meanings of ℓ and Δv .
- a. *Using the extensivity property, $U(\lambda S, \lambda V, \lambda N) = \lambda U(S, V, N)$, by making derivative with respect to λ and setting $\lambda=1$, we get the Euler equation $U = TS - pV + \mu N$. Differentiate this (i.e. $dU = \dots$) and together with the fundamental thermodynamic equation $dU = TdS - pdV + \mu dN$, we find the Gibbs-Duhem relation $SdT - Vdp + Nd\mu = 0$.*
- b. *We have two Gibbs-Duhem equations, one for the gas phase and one for liquid phase. Subtracting the two after dividing by N , and noting that in phase equilibrium on the phase coexistence line, temperature T , pressure p , and chemical potential μ must be the same. We get the Clapeyron equation, where ℓ is the latent heat needed to vaporize liquid per mole, and $\Delta v = v_g - v_l$ is the volume change per mole. All are positive quantities.*

For more details, see page 59-62, and 228-231 of the textbook of Callen, "thermodynamics and an intro to thermostatistics", 2nd ed.

2. With respect to Liouville's theorem:
 - a. Explain in words that the microcanonical ensemble is consistent with the theorem.
 - b. Let $D(t)$ be the Jacobian determinant defined by the transform $\gamma_0 = (p, q)$ to $\Gamma(t) = (p(t), q(t))$ where $\Gamma(t)$ is the solution of the Hamilton equations of motion with initial condition γ_0 . Using the integral form of Liouville's theorem, show that $D(t) = 1$.
 - c. Show that $dD(t)/dt|_{t=0} = 0$ as a consequence of the Hamilton equations of motion.
- a. *If the phase space distribution function ρ is a function of the Hamiltonian H only, then ρ satisfies the Liouville's equation $\partial \rho / \partial t = \{H, \rho\} = 0$. The microcanonical distribution, $\rho = \text{const}$ if $E < H < E + \Delta E$ and 0 otherwise is a function of H only, so it is consistent with Liouville's equation. But so does the canonical distribution. Liouville says the phase space volume is invariant under the Hamiltonian dynamics. This implies that if the density was initially uniform (constant in a region), it will stay so, so microcanonical ensemble is stationary and consistent with Liouville theorem.*

b. Let Ω_0 be a set of phase space points at initial time $t=0$. This region moves to Ω_t by time t . There is a one-to-one map from points Ω_0 to Ω_t . The Liouville theorem says (see, e.g., V.I. Arnold, “mathematical methods of classical Mechanics,” 2nd ed, page 69):

$$\int_{\Omega_0} d\gamma_0 = \int_{\Omega_t} d\Gamma_t$$

However, according to the usual rule of change of integration variables, we need

$$\int_{\Omega_t} d\Gamma_t = \int_{\Omega_0} \det \left(\frac{\partial \Gamma_t}{\partial \gamma_0} \right) d\gamma_0 = \int_{\Omega_0} D(t) d\gamma_0$$

Compare the two, we conclude that the Jacobian determinant $D(t)$ must be 1 for all t .

c. $D(t)$ is the determinant of matrix elements $\frac{\partial \Gamma_i}{\partial \gamma_j}$ (where Γ is at time t and γ at time 0).

Expand Γ for small time t using the Hamilton's equation, and taking the time derivative of the determinant and set time to 0, we obtain.

$$\left. \frac{dD(t)}{dt} \right|_{t=0} = \sum_{i=1}^{2N} \left. \frac{d}{dt} \left(\frac{\partial \Gamma_i}{\partial \gamma_i} \right) \right|_{t=0}$$

Using the solution of Hamilton's equation near $t=0$, e.g., $P_i = p_i^0 - t \partial H / \partial q_i^0 + O(t^2)$, we can show the sum above is zero due to the fact that the second order mixed derivative to

$$H \text{ are equal, where } \Gamma = \begin{pmatrix} \Gamma_1 \\ \Gamma_2 \\ \dots \\ \Gamma_{2N} \end{pmatrix} = \begin{pmatrix} P_1 \\ P_2 \\ \dots \\ P_N \\ Q_1 \\ Q_2 \\ \dots \\ Q_N \end{pmatrix} \text{ and similarly for the small } \gamma \text{ version.}$$

3. Consider a non-interacting gas of N monatomic molecules in a cylinder with horizontal cross-section area of A and vertical height L in a uniform gravitational field. The total volume is $V=AL$. In addition to the usual kinetic energy term $p^2/(2m)$, there is also a potential energy of mgz , $0 < z < L$, for each molecule of mass m . Where g is the gravitational acceleration constant.

- Give the expression for the Boltzmann entropy S in microcanonical ensemble in terms of some integral with proper integral limit or constraint (You need not perform the integral analytically).
- Determine the partition function Z of the canonical ensemble.
- What is the pressure P at bottom ($z=0$) and top ($z=L$) of the cylinder?

a. Boltzmann entropy is $S = k_B \ln W$, where

$$W = \frac{A^N}{N! h^{3N}} \int_{U < \sum_{i=1}^N \left(\frac{\mathbf{p}_i^2}{2m} + mgz_i \right) < U + \Delta} dz_1 dz_2 \dots dz_N d^3 \mathbf{p}_1 d^3 \mathbf{p}_2 \dots d^3 \mathbf{p}_N$$

where $0 < z_i < L$.

$$\begin{aligned} b. Z &= \frac{1}{N!} \int_A d^3 \mathbf{r} \int_{z_0}^{z_1} dz \int_{-\infty}^{+\infty} dp_x \int_{-\infty}^{+\infty} dp_y \int_{-\infty}^{+\infty} dp_z e^{-\beta \left(\frac{p_x^2 + p_y^2 + p_z^2}{2m} + mgz \right)} \\ &= \frac{A}{h^3} \left(\frac{2\pi m}{\beta} \right)^{3/2} \frac{e^{-\beta mgz_0} - e^{-\beta mgz_1}}{\beta mg} \end{aligned}$$

where z_0 is close to 0 and z_1 close to L . The reason that we don't set exactly to 0 and L is because we like to see how the free energy varies with the locations of the top and bottom of the cylinder so that pressure there can be calculated.

c. The pressure is the force per unit area, f/A . The work done to the system is force times the displacement. So we have $dF = f_0 dz_0 - f_1 dz_1 - SdT$, $F = -k_B T \ln Z$. Differentiate with respect to z_0 or z_1 , we get the pressure at the bottom and top as

$$P(z=0) = \frac{Nmg}{(1 - e^{-\beta mgL})A},$$

$$P(z=L) = \frac{Nmg}{(e^{\beta mgL} - 1)A}.$$

4. Consider an idealized boson gas (free particle) such that each quantum particle can take only two non-degenerate quantum state, either the ground state with energy $\varepsilon_0 = 0$, or excited state with energy ε_1 , at a chemical potential μ .
- Determine the grand partition function Ξ .
 - Determine if such a simple system shows Bose-Einstein condensation phase transition. If it does, determine the transition temperature T_c . If not, discuss why it does not have a phase transition as comparing to the usual problem of non-interacting Bose particles in a box.

a. *The grand partition function is*

$$\begin{aligned}\Xi &= \sum_{\text{All Fock states}} e^{-\beta(H_N - \mu N)} \\ &= \sum_{n_0=0,1,2,\dots, n_1=0,1,2,\dots} e^{-\beta(\varepsilon_0 n_0 + \varepsilon_1 n_1 - \mu n_0 - \mu n_1)} \\ &= \frac{1}{1 - e^{\beta\mu}} \cdot \frac{1}{1 - e^{\beta(\mu - \varepsilon_1)}}\end{aligned}$$

- b. *The occupation number of the two levels are $\langle n_0 \rangle = 1/(\exp(-\beta\mu) - 1)$, $\langle n_1 \rangle = 1/(\exp(\beta(\varepsilon_1 - \mu)) - 1)$. Thus the total $N = \langle n_0 \rangle + \langle n_1 \rangle$. This equation can be used to determine the chemical potential μ as a function of T , which is a smooth function of T . Alternatively, one can calculate heat capacity C , which is a smooth function of T again. The reason that the usual particle-in-a-box boson particles have phase transition is that there are infinite numbers of excited states. So the total number of particle is determined by a separate term for the ground state, plus an integral for all the excited states. The chemical potential is not a smooth function of T due to this.*